12TH CHEMISTRY · CHAPTER 29 · SOLVED EXERCISE

Spectroscopy–1

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS :

I) Which of the following compounds could produce the IR spectrum shown below? [Spectrum: broad band X at about 3400 cm^{-1}; strong sharp band Y at about 1700 cm^{-1}]

(a) propanoic acid (b) 4-hydroxybutanone
(c) pentan-1-ol (d) 2-hydroxybut-1-ene

Answer: (b) 4-hydroxybutanone — X (about 3200–3600 cm^{-1}) is the O–H of a hydroxy group and Y (about 1700 cm^{-1}) is C=O. Only 4-hydroxybutanone has both; a carboxylic acid would show a much broader O–H band at 2500–3300 cm^{-1}.

II. What is the wavelength range of the UV spectrum?

(a) 100 nm to 500 nm (b) 200 nm to 800 nm
(c) 300 nm to 1000 nm (d) 400 nm to 1600 nm

Answer: (b) 200 nm to 800 nm — UV-Vis spectroscopy uses wavelengths from 200–800 nm.

III. The λ of σ to σ*transitions lies in the

(a) IR region (b) Visible region
(c) UV region (d) None of the above

Answer: (c) UV region — σ → σ* needs the highest energy, so it absorbs at very short wavelength in the (far) UV region.

IV. The types of transitions possible in UV-visible region for a compound with molecular formula C_2H_4 are:

(a) n→π* (b) n→ σ*
(c) σ →σ* (d) π →π*

Answer: (d) π →π* — Ethene has a π bond and no lone pairs, so only π → π* is possible in this region.

V. What is index of hydrogen deficiency of C_8H_8?

(a) 1 (b) 2
(c) 3 (d) 4

Answer: none of the printed options is correct.

$$IHD = (2 \times 8 + 2 - 8) \div 2 = 10 \div 2 = 5$$

Textbook error: for C_8H_8 (e.g. styrene: one ring + four double bonds) the IHD is 5, which is not among the options. The nearest printed option is (d) 4, which would be correct for C_6H_6 or C_8H_{10}.

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VI. Which of the following statements about the mass spectrum of CH_3Cl is correct?

(a) There is one peak for the molecular ion with an m/e value of 45.

(b) There is one peak for the molecular ion with an m/e value of 52.

(c) The last two peaks have abundances in the ratio 1:1 and occur at m/e values of 50 and 52.

(d) The last two peaks are of equal size and occur at m/e values of 50 and 52.

Answer: none of the printed options is fully correct. CH_3Cl gives two molecular ion peaks, at m/e 50 ($CH_3{}^{35}Cl^+$, M) and m/e 52 ($CH_3{}^{37}Cl^+$, [M+2]), with heights in the ratio **3 : 1**.

Textbook error: options (c) and (d) give the right m/e values (50 and 52) but the wrong ratio (1 : 1 / equal size is the pattern of bromine). The intended answer is probably (c) with the ratio printed as 3 : 1.

Q2. SHORT ANSWER QUESTIONS

a) Bromobutane, $CH_3CH_2CH_2CH_2Br$, can be reacted with hot aqueous sodium hydroxide to prepare butan-1-ol.
$CH_3CH_2CH_2CH_2Br + OH^- \rightarrow CH_3CH_2CH_2CH_2OH + Br^-$
Butan-1-ol produced can be analysed by mass spectrometry.
i. Predict the formula of fragment ion which is produced to m/e value of 17.
ii. Predict two other fragment ions that you would expect to see in the mass spectrum of butan-1-ol and state the m/e value of each ion.

i. OH^+ (16 + 1 = 17).

ii. Any two of: $CH_3{}^+$ (m/e 15), $C_2H_5{}^+$ (m/e 29), CH_2OH^+ (m/e 31), $C_3H_7{}^+$ (m/e 43), $C_4H_9{}^+$ (m/e 57). (The molecular ion $C_4H_9OH^+$ is at m/e 74.)

b) Identify the functional groups responsible for the peaks labelled X and Y.

X (broad, about 3200–3600 cm^{-1}): the O–H bond of a hydroxy (alcohol) group. Y (strong, sharp, about 1700 cm^{-1}): the C=O bond of a carbonyl group.

c) Explain how the two isomers shown in the diagram can be distinguished using IR spectroscopy. [A: cyclopentanone; B: cyclopent-2-en-1-ol]

A (ketone) shows a strong, sharp C=O absorption at about 1670–1740 cm^{-1} and no O–H band. B (unsaturated alcohol) shows a strong, broad O–H absorption at 3200–3600 cm^{-1} and a C=C absorption at 1500–1680 cm^{-1}, but no C=O peak.

d) The molecular formulas of but-1-ene and but-1,4-diene are given below:
but-1-ene: C_4H_8 but-1,3-diene: C_4H_6
i. Fine out the IHD for both of these compounds.
ii. Compare the UV absorption spectrum of but-1-ene to that of but-1,3-diene by stating the difference in their λ_{max} qualitatively.

The stem prints "but-1,4-diene", which cannot exist; the formula line correctly gives but-1,3-diene.

i. but-1-ene, C_4H_8: IHD = (10 − 8) ÷ 2 = 1

but-1,3-diene, C_4H_6: IHD = (10 − 6) ÷ 2 = 2

ii. But-1,3-diene has two conjugated double bonds, so the energy gap for the π → π* transition is smaller and it absorbs at a longer wavelength (λ_{max} = 217 nm). But-1-ene has a single, unconjugated double bond; its π → π* transition needs more energy, so its λ_{max} is at a shorter wavelength (below 200 nm).

Q3. CONSTRUCTED RESPONSE QUESTIONS

a) Compound X is an atmospheric pollutant emitted from fuel combustion of petrol and diesel vehicles. Compound X is a potent human carcinogen. Analysis of compound X showed the following percentage composition by mass: C, 88.89% and H, 11.1%. Mass spectrometry showed a molecular ion peak at m/e = 54. Compound X reacts with H_2 in the presence of a nickel catalyst in a 1 : 2 molar ratio. Analyse and interpret this information to determine a possible structure for compound X. Show all your working.

Step 1: Empirical formula

C: 88.89 ÷ 12 = 7.41 H: 11.1 ÷ 1 = 11.1

Ratio C : H = 1 : 1.5 = 2 : 3 → empirical formula C_2H_3 (mass 27)

Step 2: Molecular formula

Molecular mass = 54 (molecular ion peak); 54 ÷ 27 = 2

Molecular formula = C_4H_6

Step 3: IHD

IHD = (2 × 4 + 2 − 6) ÷ 2 = 2

Step 4: Structure

One mole of X adds two moles of H_2, so both degrees of unsaturation are π bonds (two C=C bonds or one C≡C bond) and there is no ring. A possible structure is **buta-1,3-diene, CH_2=CH–CH=CH_2**.

$$CH_2=CH–CH=CH_2 + 2H_2 \rightarrow CH_3CH_2CH_2CH_3 \text{ (Ni)}$$

b) How will you differentiate between the following using the UV visible spectroscopy. [i) a cyclohexene ring whose C=C is conjugated with a diene side chain; ii) a cyclohexane ring carrying the same diene side chain]

Compound (i) has **three** conjugated double bonds (the ring C=C is conjugated with the two C=C bonds of the side chain), while compound (ii) has only **two** conjugated double bonds. More conjugation means a smaller energy gap, so (i) absorbs at a longer wavelength: its λ_{max} is about 30–40 nm higher than that of (ii). (Compare buta-1,3-diene 217 nm and hexa-1,3,5-triene 257 nm.)

DESCRIPTIVE QUESTIONS

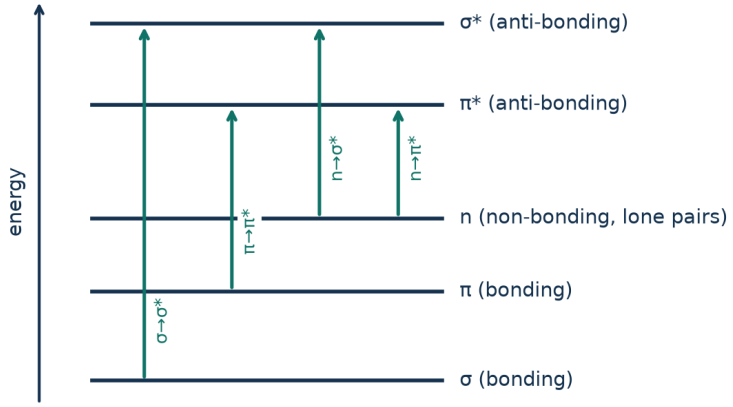
Q4. Describe various possible electronic transitions in a compound having sigma, pi bonds, and non-bonding electrons.

A compound may contain σ electrons (σ-bonds), π electrons (in unsaturated compounds only) and n electrons (non-bonded electrons). σ bonding orbitals are lower in energy than π bonding orbitals, which are lower than non-bonding orbitals. The anti-bonding orbitals σ* and π* are normally empty.

When radiation of the correct frequency is absorbed, an electron is excited from a filled orbital into an empty anti-bonding orbital. The important transitions are:

- π → π* (from pi bonding to pi anti-bonding orbitals)
- n → π* (from non-bonding to pi anti-bonding orbitals)
- n → σ* (from non-bonding to sigma anti-bonding orbitals)

σ → σ* needs the most energy and lies outside the 200–800 nm range. n → π* needs the least energy.



Electronic transitions from filled orbitals to anti-bonding orbitals

Q5. Explain two types of atomic spectrum.

Atomic spectra are observed when atoms emit or absorb light of certain wavelengths. They provide evidence that electrons in atoms can only move between discrete energy levels.

Emission spectrum: it is produced by heating a low-pressure gas. Heating excites electrons to higher energy levels. When an electron falls back to a lower level it emits a photon; each transition corresponds to a specific wavelength, seen as a spectral line. The emission spectrum is a set of **coloured lines on a black background**.

Absorption spectrum: it is produced by passing white light through a cool, low-pressure gas. Only photons with the exact energy needed to excite the ground-state electrons are absorbed. Each absorbed wavelength appears as a **dark line on a coloured background**. These lines are at the same positions as the lines in the

emission spectrum of the same element, so an absorption spectrum is like a photographic negative of an emission spectrum.

Q6. How mass spectrometry is used to determine the isotopic mass and average atomic masses of elements? Explain giving an example.

Mass spectrometry is an analytical technique used to identify unknown compounds. The molecules or atoms of a small sample are bombarded with high-energy electrons, which makes them lose an electron. The positive ion formed is detected and recorded as m/e (mass to charge ratio) against percentage abundance.

Mass spectrometry finds the relative abundance of isotopes experimentally. Relative abundance is the proportion of a particular isotope in the natural mixture of isotopes. The heights of the peaks show the proportion of each isotope. For example, Cl-35 and Cl-37 are 75% and 25%; boron-10 is 19.9% and boron-11 is 80.1%.

$$\text{Relative atomic mass (Ar)} = \Sigma \text{ (isotopic mass} \times \text{\% abundance)} \div 100$$

Example: the mass spectrum of boron shows two peaks, at m/e 10 (19.9%) and m/e 11 (80.1%).

$$\text{Ar of boron} = [(19.9 \times 10) + (80.1 \times 11)] \div 100$$
$$= 10.801 = 10.80 \text{ (to 2 decimal places)}$$

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

Quick Check 29.1

a) Calculate the IHD for each of the following. i) C_6H_6 ii) C_6H_{12} iii) C_4H_4

For a hydrocarbon CnHm: $\text{IHD} = (2n + 2 - m) \div 2$

i) C_6H_6: $\text{IHD} = (14 - 6) \div 2 = 4$

ii) C_6H_{12}: $\text{IHD} = (14 - 12) \div 2 = 1$

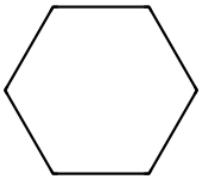
iii) C_4H_4: $\text{IHD} = (10 - 4) \div 2 = 3$

b) Propose the structure of the above compounds.

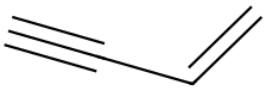
One possible structure for each: (i) benzene – one ring and three double bonds (IHD 4); (ii) cyclohexane – one ring (or hex-1-ene, one double bond); (iii) but-1-en-3-yne, CH_2=CH–C≡CH – one double and one triple bond (IHD 3).



(i) benzene



(ii) cyclohexane



(iii) but-1-en-3-yne

c) Hopanes are the waxy solids found in petroleum products as residues after refining the crude oil. A hopane has the formula $C_{30}H_{62}$, i) Find its IHD. ii) Predict whether this compound is saturated or not.

$$\text{i) IHD} = (2 \times 30 + 2 - 62) \div 2 = 0$$

ii) IHD = 0, so the compound (as printed) is saturated: it has no rings and no multiple bonds.

Quick Check 29.2

a) Which type of electronic transitions require least energy in UV/Vis spectroscopy? Explain. i) σ→σ* ii) π→π* iii) n→σ* iv) n→π*

iv) n → π*. The non-bonding (n) orbital is the highest-energy filled orbital and π* is the lowest-energy empty orbital, so the gap between them is the smallest.

b) Arrange the following in an increasing order of λ_{max}. i) a ketone whose C=O is conjugated with two C=C bonds (with an ethyl branch) ii) but-3-en-2-one, CH_2=CH–CO–CH_3 iii) hexa-3,5-dien-2-one, CH_2=CH–CH=CH–CO–CH_3

ii < iii < i. λ_{max} increases with the extent of conjugation. (ii) has only one C=C conjugated with C=O, so it absorbs at the shortest wavelength. (iii) and (i) both have two C=C bonds conjugated with C=O; (i) also carries an extra alkyl group on the conjugated chain, which shifts λ_{max} a little further.

The effect of the alkyl branch (placing i above iii) is beyond the textbook; from the chapter alone, i and iii have the same extent of conjugation and both absorb at a longer wavelength than ii.

c) Will $CH_3CHOHCH_3$ show a significant absorption in the UV-Vis region.

No. Propan-2-ol has only σ bonds and lone pairs on oxygen; it has no π electrons. Its possible transitions (σ → σ* and n → σ*) need high energy and fall below 200 nm, outside the UV-Vis range (200–800 nm).

Quick Check 29.2 (second box, infrared)

The textbook numbers two different boxes "Quick Check 29.2"; they are solved in the printed order.

I. Which one of the infra-red spectra is that of butanone and which one is of butan-2-ol?

Spectrum A is butanone: it has a strong, sharp absorption near 1700 cm^{-1} (C=O) and no broad O–H band. **Spectrum B is butan-2-ol:** it has a strong, broad absorption at about 3200–3600 cm^{-1} (O–H) and no C=O peak.

ii. Compounds T and U are isomers with the molecular formula $C_3H_6O_2$. Suggest their structures based on the spectra shown below:

T is propanoic acid, CH_3CH_2COOH: its spectrum shows a very broad O–H absorption (about 2500–3300 cm^{-1}) together with a strong C=O absorption near 1700 cm^{-1}. **U is an ester, methyl ethanoate CH_3COOCH_3** (ethyl methanoate, $HCOOCH_2CH_3$, is also possible): its spectrum has a strong C=O absorption (1710–1750 cm^{-1}) and C–O absorption but no O–H band.

Quick Check 29.3

a) Show all the possible fragments of i) C_4H_{10} ii) C_5H_{12} iii) C_7H_{16}

Taking straight-chain alkanes, the C–C bonds break to give alkyl ions:

Compound	Molecular ion (m/e)	Fragment ions (m/e)
i) C_4H_{10}	$C_4H_{10}^+$ (58)	$C_3H_7^+$ (43), $C_2H_5^+$ (29), CH_3^+ (15)
ii) C_5H_{12}	$C_5H_{12}^+$ (72)	$C_4H_9^+$ (57), $C_3H_7^+$ (43), $C_2H_5^+$ (29), CH_3^+ (15)
iii) C_7H_{16}	$C_7H_{16}^+$ (100)	$C_6H_{13}^+$ (85), $C_5H_{11}^+$ (71), $C_4H_9^+$ (57), $C_3H_7^+$ (43), $C_2H_5^+$ (29), CH_3^+ (15)

b) The mass spectrum of CH_3Br is given below. [Peaks at m/e 15, 94 and 96] i) Identify the fragment against each peak. ii) What is the justification of two peaks at 94 and 96?

i) m/e 15: CH_3^+; m/e 94: $CH_3{}^{79}Br^+$ (molecular ion, M); m/e 96: $CH_3{}^{81}Br^+$ ([M+2] peak).

ii) Bromine exists as two isotopes, ^{79}Br and ^{81}Br, in almost equal abundance. So there are two molecular ion peaks, M and [M+2], two units apart and of similar height (1 : 1).

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